Case Study 18: Fuel Consumption and Route Comparison

1. Problem Statement

Calculate fuel requirements and journey costs for several vehicles across multiple candidate routes using distance, mileage, fuel prices, and mandatory safety reserve margins, and identify the lowest-cost feasible route-vehicle combination.

2. Objectives

- Calculate total required fuel volume including safety reserve requirements.
- Compute total monetary journey cost per vehicle-route combination.
- Identify feasible routes within vehicle fuel tank capacities and evaluate the most cost-effective option.

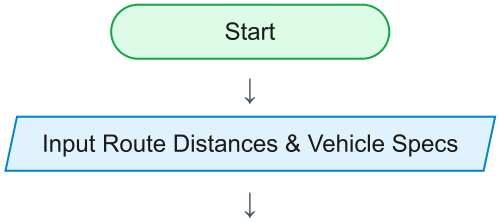
3. Assumptions

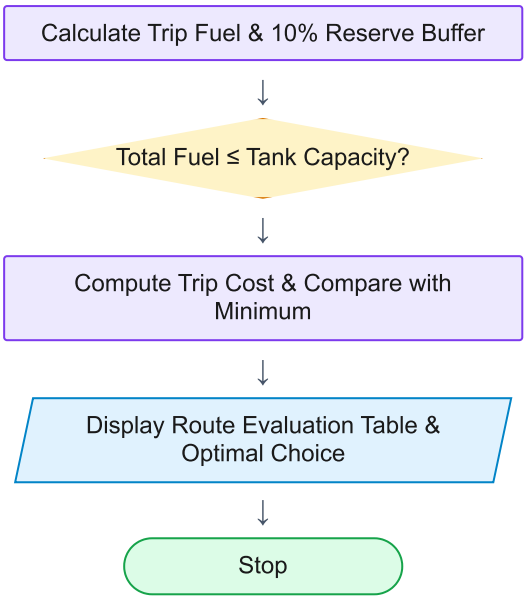
- **Fuel Reserve Margin:** Mandatory extra **10% fuel reserve** required for all routes.
- **Fuel Price:** Petrol/Diesel price is set to **Rs. 95.00 per liter**.
- Vehicles with total required fuel volume exceeding tank capacity are flagged as *Infeasible* unless refuel stops are made.

4. Algorithm

1. Start.
2. Read route distances and vehicle parameters (mileage km/L, fuel capacity L).
3. For each route and vehicle combination:
   - Calculate trip fuel needed = `distance / mileage`.
   - Calculate total fuel needed with safety reserve = `trip_fuel * 1.10`.
   - Calculate total journey cost = `total_fuel * fuel_price`.
   - Check feasibility: `total_fuel <= tank_capacity`.
4. Track the minimum journey cost among all feasible combinations.
5. Display comparative analysis report and highlight the optimal selection.
6. Stop.

5. Flowchart





6. Pseudocode

```
START INPUT routes (name, distance) INPUT vehicles (name, mileage, tank_capacity) INPUT fuel_price = 95.00 best_cost = INFINITY best_option = NONE FOR EACH route IN routes DO FOR EACH vehicle IN vehicles DO trip_fuel = route.distance / vehicle.mileage total_fuel = trip_fuel * 1.10 cost = total_fuel * fuel_price IF total_fuel <= vehicle.tank_capacity THEN status = "FEASIBLE" IF cost < best_cost THEN best_cost = cost best_option = (route.name, vehicle.name) ENDIF ELSE status = "INFEASIBLE" ENDIF PRINT route.name, vehicle.name, total_fuel, cost, status ENDFOR ENDFOR PRINT best_option, best_cost STOP
```

7. Complete Python Source Code

```
      # Fuel Consumption and Route Comparison def
evaluate_routes_and_vehicles(routes, vehicles, fuel_price=95.00): reserve_factor =
1.10 # 10% safety reserve margin print("=" * 70) print(" FUEL CONSUMPTION & ROUTE
COMPARISON REPORT") print("=" * 70) print(f"{'Route':<10}{'Vehicle':<12}
{'Dist(km)':<10}{'Fuel(L)':<10}{'Cost (Rs.)':<14}{'Feasible?':<10}") print("-" *
70) best_cost = float('inf') optimal_choice = None for r_name, dist in
routes.items(): for v_name, specs in vehicles.items(): mileage = specs["mileage"]
capacity = specs["capacity"] trip_fuel = dist / mileage total_fuel = trip_fuel *
reserve_factor cost = total_fuel * fuel_price is_feasible = total_fuel <= capacity
status = "YES" if is_feasible else "NO (Refuel)" if is_feasible and cost <
best_cost: best_cost = cost optimal_choice = (r_name, v_name, dist, total_fuel)
print(f"{r_name:<10}{v_name:<12}{dist:<10}{total_fuel:<10.2f}{cost:<14,.2f}{status:
<10}") print("-" * 70) if optimal_choice: r, v, d, f = optimal_choice
print(f"OPTIMAL FEASIBLE SELECTION: Route {r} using {v}") print(f" └─ Distance: {d}
km | Fuel Needed (inc. 10% reserve): {f:.2f} L") print(f" └─ Minimum Total Journey
Cost: Rs. {best_cost:,.2f}") else: print("No feasible single-tank options
available.") print("=" * 70) # Test Data routes_data = { "Route A": 320, # Highway
route "Route B": 280, # Short but hilly route "Route C": 350 # Bypass route }
vehicles_data = { "Hatchback": {"mileage": 18.0, "capacity": 35.0}, "Sedan":
{"mileage": 14.0, "capacity": 45.0}, "SUV": {"mileage": 10.0, "capacity": 50.0} }
evaluate_routes_and_vehicles(routes_data, vehicles_data)
```

8. Test Case and Expected Output

Evaluation Output for Test Scenario:

```
      ====================================================================== FUEL
CONSUMPTION & ROUTE COMPARISON REPORT
====================================================================== Route
Vehicle Dist(km) Fuel(L) Cost (Rs.) Feasible? ----------------------------------
-------------------------------- Route A Hatchback 320 19.56 1,857.78 YES Route A
Sedan 320 25.14 2,388.57 YES Route A SUV 320 35.20 3,344.00 YES Route B Hatchback
280 17.11 1,625.56 YES Route B Sedan 280 22.00 2,090.00 YES Route B SUV 280 30.80
2,926.00 YES Route C Hatchback 350 21.39 2,031.94 YES Route C Sedan 350 27.50
2,612.50 YES Route C SUV 350 38.50 3,657.50 YES ----------------------------------
-------------------------------- OPTIMAL FEASIBLE SELECTION: Route Route B using
Hatchback └─ Distance: 280 km | Fuel Needed (inc. 10% reserve): 17.11 L └─ Minimum
Total Journey Cost: Rs. 1,625.56
======================================================================
```

9. Conclusion

The Fuel Consumption and Route Comparison model reliably evaluates journey costs with built-in safety margins and tank capacity constraints, helping travelers choose the most economical vehicle-route combination.